

Linear Algebra

How to calculate subject grades

- ① Midterm exam 30%: 18 multiple choice questions - 50 minutes
- ② Major assignment 30%: 20% teamwork and 10% assignment on LMS.hcmut.edu.vn
- ③ Final Exam 40%: 22 multiple choice questions (70 minutes).

Attention:

- Incorrectly selected multiple-choice questions will result in a deduction of 1/5 of the point for that question; unanswered questions will not result in a deduction.
- A score of 0 in any component or absence from the exam will result in failing.
- Final exam scores ≤ 2 will result in failure.



The content of the course includes the following chapters

- 1) Complex numbers
- 2) Matrix – Determinant – System of Equations
- 3) Vector Space
- 4) Euclidean Space
- 5) Linear Mapping
- 6) Eigenvalue – Eigenvector



Reference Materials

- Textbook: Đại số tuyến tính - Đặng Văn Vinh
- Đỗ Công Khanh. Đại số tuyến tính. NXB Đại học quốc gia
- Lay, David C., Steven R. Lay, and Judi J. McDonald. Linear Algebra and its applications 5th edition. Pearson 2016.
- Elementary Linear Algebra with Applications - Howard Anton
- Matrix analysis and applied linear algebra- Carl D Meyer



Chapter 1. Complex Numbers

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Content

- 1 Basic concepts of complex numbers
- 2 Polar form
- 3 Exercises



1 Basic concepts of complex numbers

2 Polar form

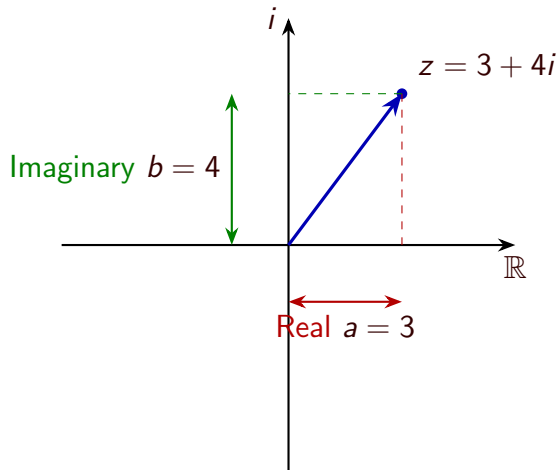
3 Exercises

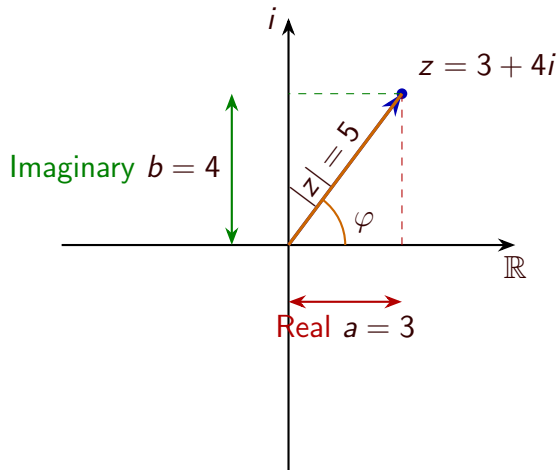


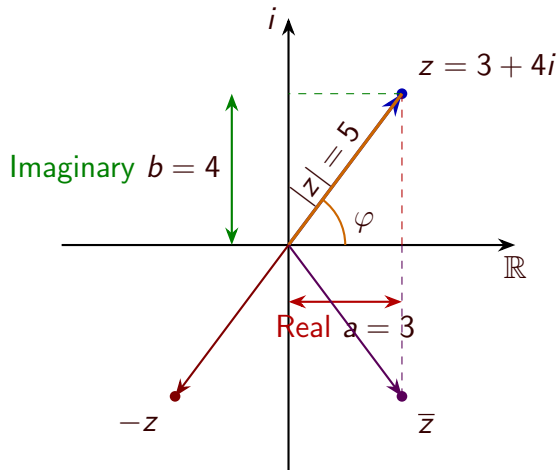
Basic concepts of the algebraic form of complex numbers

- **Imaginary unit:** $i^2 = -1$
- **Complex number:** $z = a + bi$ với $a, b \in \mathbb{R}$
- **Real part:** $Re(z) = a$
- **Imaginary part:** $Im(z) = b$
- **Pure imaginary number:** $a = 0$ ($z = bi$)
- **Conjugate:** $\bar{z} = a - bi$
- **Modulus:** $|z| = \sqrt{a^2 + b^2}$









Complex number operations

- **Addition:** $(a + bi) + (c + di) = (a + c) + (b + d)i$
- **Subtraction:** $(a + bi) - (c + di) = (a - c) + (b - d)i$
- **Multiplication:** $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$
- **Division:** $\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2}$

Addition, subtraction, multiplication, and division of complex numbers are similar to real numbers, with some important considerations.

$$i^2 = -1, i^3 = \frac{1}{i} = -i, i^4 = 1 \dots$$



Properties of the modulus and related operations

- $|z_1 z_2| = |z_1| |z_2|$
- $|z^k| = |z|^k, \forall k \in \mathbb{Z}, z \neq 0$
- $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ (với $z_2 \neq 0$)
- $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle inequality)
- $|\bar{z}| = |z|$
- $z\bar{z} = |z|^2 = |\bar{z}|^2$



Example 1: Find the real part, imaginary part, modulus, and the conjugate of the following complex numbers.

a. $z = (1 + i)(2 - i).$

b. $z = (2 - i)^2.$

c. $z = \frac{1 + i}{3 - 4i}$

d. $z = \frac{1 - 2i}{3i - 1} + (2 - 3i)^2$



Example 2

Let $z = \frac{(1 - 2i)i^{2025}}{3i - 1}$

- a. Find the modulus of z
- b. Simplify z , then find its conjugate, real part, and imaginary part.



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- Find the modulus of z
- Simplify z , then find its conjugate, real part, and imaginary part.

Solution:

- $i^{2025} = i^{4 \times 506 + 1} = i$
- $z = \frac{(1 - 2i)i}{3i - 1} = \frac{i + 2}{3i - 1}$
- Multiply the numerator and denominator by the conjugate. : $z = \frac{(i + 2)(-3i - 1)}{(3i - 1)(-3i - 1)}$



Example 2 (cont.)

Calculate:

$$\begin{aligned} z &= \frac{(i+2)(-3i-1)}{9+1} = \frac{1}{10}[(2+i)(-1-3i)] \\ &= \frac{1}{10}(-2-6i-i-3i^2) = \frac{1}{10}(1-7i) \\ &= \frac{1}{10} - \frac{7}{10}i \end{aligned}$$

$$\text{a. } |z| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(\frac{7}{10}\right)^2} = \frac{\sqrt{50}}{10} = \frac{\sqrt{2}}{2}$$

$$\text{b. } \bar{z} = \frac{1}{10} + \frac{7}{10}i, \operatorname{Re}(z) = \frac{1}{10}, \operatorname{Im}(z) = -\frac{7}{10}$$



Example 3

Let $z = (1 + 2i)^2(a + 2i)$, $a \in \mathbb{R}$

- a. Find a so that z is a real number.
- b. Find a so that z is a purely imaginary number.



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Solution:

$$\begin{aligned}
 (1 + 2i)^2(a + 2i) &= (1 + 4i + (2i)^2)(a + 2i) = (1 + 4i - 4)(a + 2i) = (-3 + 4i)(a + 2i) \\
 &= (-3)(a + 2i) + (4i)(a + 2i) = -3a - 6i + 4ai + 8i^2 = -3a - 6i + 4a - 8 \\
 &= (-3a - 8) + (4a - 6)i.
 \end{aligned}$$



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a. z is a real number if $4a - 6 = 0 \iff a = \frac{3}{2}$

b. z is a purely imaginary number if $-3a - 8 = 0 \iff a = -\frac{8}{3}$



Example 4. Find z so that

a. $z^2 = 3 - 4i$

b. $z^2 = 5 - 12i$

c. $z^2 = 7 - 24i$



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Solution

a. Suppose $z = x + yi$

$$z^2 = (x + yi)^2 = x^2 - y^2 + 2xyi. \Rightarrow \begin{cases} x^2 - y^2 = 3, \\ 2xy = -4 \Rightarrow xy = -2. \end{cases}$$

From $xy = -2 \Rightarrow y = -\frac{2}{x}$. Substitute $x^2 - \frac{4}{x^2} = 3 \Rightarrow x^4 - 3x^2 - 4 = 0$.

$$(x^2 - 4)(x^2 + 1) = 0 \Rightarrow x^2 = 4 \Rightarrow \begin{cases} x = 2, y = -1 \Rightarrow z = 2 - i \\ x = -2, y = 1 \Rightarrow z = -2 + i \end{cases}$$



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b. $z = \pm(3 - 2i)$.



Example 5: Solve the equation

a. $z^2 - 4z + 13 = 0$

b. $z^2 + (1 - i)z - 5i = 0$

c. $z^2 + (1 - 2i)z + (3 + 4i) = 0$



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a. $\Delta = (-4)^2 - 4 \cdot 1 \cdot 13 = 16 - 52 = -36 = (6i)^2.$

It follows that $z = \frac{4 \pm 6i}{2} = 2 \pm 3i$. Hence $z = 2 + 3i$ or $z = 2 - 3i$.



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b. $\Delta = (1 - i)^2 - 4 \cdot 1 \cdot (-5i) = 18i = (3 + 3i)^2.$

It follows that $z = \frac{-(1 - i) \pm (3 + 3i)}{2}$. Therefore $\boxed{z = 1 + 2i \text{ or } z = -2 - i.}$

c. $\boxed{z_1 = \frac{-1 + 2i + \sqrt{5}(1 - 2i)}{2} \text{ or } z_2 = \frac{-1 + 2i - \sqrt{5}(1 - 2i)}{2}.}$



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Polar form of a complex number

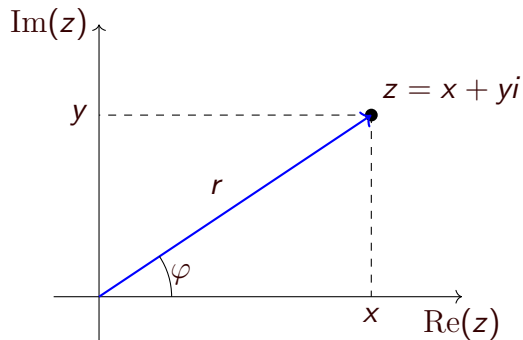
Geometric form

- **Polar form:** $z = r(\cos \varphi + i \sin \varphi)$
- **Exponential form:** Set $e^{i\varphi} = \cos \varphi + i \sin \varphi$

$$z = re^{i\varphi}$$

- **Argument:** $\arg(z)$ - the angle between the vector representing z and the positive real axis.
- Với $r = |z|$, $\varphi = \arg(z)$
- **Principal argument:** $\text{Arg}(z) \in (-\pi, \pi]$ hoặc $\text{Arg}(z) \in [0, 2\pi)$.

Geometric representation of a complex number



- $z = r(\cos \varphi + i \sin \varphi)$.
- $\begin{cases} x = r \cos \varphi, \\ y = r \sin \varphi \end{cases}$
- $r = \sqrt{x^2 + y^2}$
- $\tan \varphi = \frac{y}{x}$ where

$$\varphi = \begin{cases} \text{acute angle,} & \text{if } x > 0, \\ \text{obtuse angle,} & \text{if } x < 0 \end{cases}$$



Some properties of argument

- Multiplication: $z_1 z_2 = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$
- Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\varphi_1 - \varphi_2)}$
- Exponentiation: $z^n = r^n e^{in\varphi}, \forall n \in \mathbb{Z}, z \neq 0.$ (Moivre fomular)



Example 4: Find modulus and Principal Argument of the following complex numbers

a. $z = 1 - i\sqrt{3}$

b. $z = \frac{-1 - i}{i\sqrt{12} - 2} i^{2025}$

c. $z = \frac{(i - \sqrt{3})^7}{w^3}$, given $\arg(w) = \frac{\pi}{7}$ and $|w| = 4$



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Solve a:

$$|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\tan \varphi = -\sqrt{3}, \quad \varphi \text{ in the fourth quadrant.}$$

$$\text{Arg}(z) = -\frac{\pi}{3} \text{ or } \frac{5\pi}{3}$$



Example 4 (cont.)

Solve b: $|z| = \frac{|-1-i|}{|i\sqrt{12}-2|} |i|^{2025} = \frac{\sqrt{2}}{4} \cdot 1^{2025} = \frac{\sqrt{2}}{4}.$

We have $i^{2025} = i^{4 \cdot 506 + 1} = i^1 = i.$

$$\begin{aligned} \arg(z) &= \operatorname{Arg}(-1-i) - \operatorname{Arg}(i\sqrt{12}-2) + \operatorname{Arg}(i) + k2\pi \\ &= \frac{5\pi}{4} - \frac{2\pi}{3} + \frac{\pi}{2} + k2\pi = \frac{13\pi}{12} + k2\pi \Rightarrow \boxed{\operatorname{Arg}(z) = \frac{13\pi}{12}} \end{aligned}$$



Example 4 (cont.)

Solve b: $|z| = \frac{|-1-i|}{|i\sqrt{12}-2|} |i|^{2025} = \frac{\sqrt{2}}{4} \cdot 1^{2025} = \frac{\sqrt{2}}{4}.$

We have $i^{2025} = i^{4 \cdot 506 + 1} = i^1 = i.$

$$\begin{aligned} \arg(z) &= \text{Arg}(-1-i) - \text{Arg}(i\sqrt{12}-2) + \text{Arg}(i) + k2\pi \\ &= \frac{5\pi}{4} - \frac{2\pi}{3} + \frac{\pi}{2} + k2\pi = \frac{13\pi}{12} + k2\pi \Rightarrow \boxed{\text{Arg}(z) = \frac{13\pi}{12}} \end{aligned}$$

Solve c: $|z| = \frac{|i-\sqrt{3}|^7}{|w|^3} = \frac{2^7}{4^3} = 2.$

$$\begin{aligned} \arg(z) &= 7 \cdot \arg(i-\sqrt{3}) - 3 \cdot \text{Arg}(w) + k2\pi = 7 \cdot \frac{5\pi}{6} - 3 \cdot \frac{\pi}{7} + k2\pi \\ &= \frac{227\pi}{42} + k2\pi = 6\pi - \frac{25\pi}{42} + k2\pi \Rightarrow \boxed{\text{Arg}(z) = -\frac{25\pi}{42}} \end{aligned}$$



Example 5. Simplify

a. $z = (2 + 3i)(1 - 4i).$

Answer: $14 - 5i$

b. $z = (3 - 2i)^2.$

Answer: $5 - 12i$



Example 6.

Let $z = \frac{(2 - i)i^{101}}{2 + i}$

a. Find modulus of z .

Answer: $\frac{\sqrt{5}}{\sqrt{5}} = 1$

b. Simplify z , find conjugate, real and imaginary part.

Answer: $z = \frac{4}{5} + \frac{3}{5}i, \quad \bar{z} = \frac{4}{5} - \frac{3}{5}i.$



Example 7.

Let $z = (2 + i)^2(a - 3i)$, $a \in \mathbb{R}$.

- a. Find a so that z is a real number.
- b. Find a so that z is a purely imaginary number.

Answer: $a = -1$

Answer: $a = 3$



Example 8

a. $z^2 = 7 - 24i.$

b. $z^2 = -15 + 8i.$

Answer: $z = \pm(4 - 3i)$

Answer: $z = \pm(1 + 4i)$



Example 9.

a. $z^2 - 2z + 5 = 0.$

b. $z^2 + (2 - 3i)z - 1 - 3i = 0.$

Answer: $z = 1 \pm 2i$

Answer: $-1 + i$



Example 10. Find modulus and Arg of the complex numbers

a. $z = -2 + 2\sqrt{3}i$.

Answer: $|z| = 4, \text{Arg}(z) = \frac{2\pi}{3}$

b. $z = 1 + i\sqrt{3}$.

Answer: $|z| = 2, \text{Arg}(z) = \frac{\pi}{3}$

c. $z = (1 + 2i)(3 - i)$

Answer: $|z| = 5\sqrt{2}, \text{Arg}(z) = \frac{\pi}{4}$

d. $z = \frac{(-1+i)^7}{(2-i\sqrt{12})^3} \cdot i^{1957}$.

Answer: $|z| = \frac{\sqrt{2}}{8}, \text{Arg}(z) = \frac{5\pi}{4}$



Example 11. Find z so that

a. $(1 + 2i)z + (3 - 4i)\bar{z} = 14 - 2i$

b. $(2 + i)z + (1 - 3i)\bar{z} = 5 + 7i$

c. $(3 - 2i)z + (4 + i)\bar{z} = 20$

d. $(1 - 4i)z + (2 + 3i)\bar{z} = 7 - 5i$

Answer: $z = 2 - i$.

Answer: $z = \frac{-33 - 31i}{5}$.

Answer: $z = 2 + i$.

Answer: $z = 1 - i$.



THANK YOU FOR ATTENTION

